

CS-515 Advanced CS Practicum

Lab 4: Virtual Networking

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In this assignment, you will learn how to create virtual network containers on a Linux OS. This is the same technology used by Docker. The assignment uses available mechanisms in a standalone Linux kernel, you should not need to download or install anything. You can create arbitrary mesh networks connecting the containers and can set link parameters including capacity, delay and packet loss. To simplify your work, write shell/Perl/Python scripts to setup the network namespaces, to run experiments and to analyse the data. All the required steps (including the commands and examples) are described in the reference [1].

Exercise 4.1

1. Create two network namespaces, say `NetNsA` and `NetNsB`. In each of these, create one network interface. Experiment with the following:
 - a) Run `ping` between `NetNsA` and `NetNsB`.
 - b) Add a queue discipline with fixed delay of say 50 ms, and run `ping`.
 - c) Add a variable delay of 50ms and run `ping`.Compare a), b) and c).
2. Use `Wireshark` to monitor traffic between the namespaces. Run `ping` from one container to the other. Start `sshd` in one namespace and setup password less `ssh` between the two network containers. Connect between the containers via `ssh`. Execute a few shell commands. Observe the traffic shown in `Wireshark`.

Evaluation:

You must show the output for each of the above tasks. Describe the inferences made from the `Wireshark` capture.

Exercise 4.2

Before working on this exercise, you will need to familiarize yourself with a few useful definitions.

Definitions:

- i. A **flow** in a network is a sequence of related packets from one source end-point to one destination end-point. A flow may carry one or more files over a single TCP connection or a sequence of TCP connections. There is no concurrence in one flow.
- ii. An **elephant flow** transfers a large amount of data, typically MB to GB in size. The metric of interest is high throughput. A network has relatively few elephant flows.
- iii. **Mice flows** transfer a small amount of data, typically a few B to a few KB in size. The metric of interest is low delay. A network has large numbers of mice flows.

In this exercise, you will investigate the performance of flows between 2 hosts. The 2 hosts may be 2 network namespaces on a single laptop/PC (created above). Use `scp` or `ftp` to generate the following

- (a) elephant flow
- (b) mice flow

Use `Wireshark` to monitor the traffic. Compare (a) and (b).

Evaluation:

You must show the output for each of the above tasks. Describe the inferences made from the `Wireshark` capture.

Optional:

Use the setup created above to understand the impact of mice flows on the throughput observed by elephant flows in the network using the following steps:

- 1) Write a script *elephant* that transfers a very large file between the two hosts using FTP. If necessary, the script repeats the transfer back-to-back so that the total duration, T_{expt} is tens of seconds.
- 2) Write a script *mouse* that transfers a very small file between the two hosts using `scp`. The script repeats the transfer with a random inter-file gap drawn from an exponential distribution so that the total duration is T_{expt} . Each script records the start and end times and the file size of each transfer.
- 3) Write a control script that runs the elephant script and N_m mouse scripts in parallel. Repeat for several values of N_m . Plot elephant throughput X_e vs. N_m for different mouse file sizes.
- 4) Submit your findings in the form of a report with the codes attached. You will receive extra credit on the basis of correctness and completeness of your findings for the optional exercise.

References:

[1] Anoushka Banerjee & Shaifu Gupta, “CS549: Quick Guide to Virtual Networking”, IIT Mandi, Apr 2020. Available: <https://students.iitmandi.ac.in/moodle/>